

# Supplementary Material

## Predicting Family Educational Climate from Naturalistic Audio Sensing of Parent-Child Homework Interactions

### S1. Measure overview

The main paper modeled nine questionnaire dimensions separately. Table S1 summarizes only the measures and dimensions used in the final family-level prediction dataset.

**Table S1. Questionnaire measures corresponding to the nine modeled outcomes.**

| Instrument                         | Dimension                           | Scored items | $\alpha$ | Response scale                                                | Source                                       |
|------------------------------------|-------------------------------------|--------------|----------|---------------------------------------------------------------|----------------------------------------------|
| s-EMBU                             | Emotional warmth                    |              | .79      | 5-point scale:<br>从不 — Never                                  | Arrindell et al. (1999); Jiang et al. (2010) |
|                                    | Overprotection                      |              | .73      | 较少 — Rarely<br>一般 — Sometimes<br>经常 — Often                   |                                              |
|                                    | Rejection                           |              | .84      | 非常频繁 — Very often                                             |                                              |
| PERS-C                             | Emotional attention                 |              | .82      |                                                               | Pereira et al. (2017); Lu et al. (2022)      |
|                                    | Emotional avoidance                 |              | .65      | 5-point scale:<br>不同意 — Disagree<br>比较不同意 — Somewhat disagree |                                              |
|                                    | Emotional acceptance                |              | .49      | 中立 — Neutral<br>同意 — Agree<br>非常同意 — Strongly agree           |                                              |
|                                    | Emotional control                   |              | .65      |                                                               |                                              |
| Parent-child communication quality | Relationship-oriented communication |              | .86      | 5-point scale:<br>不同意 — Disagree<br>比较不同意 — Somewhat disagree | Chi (2011)                                   |
|                                    | Problem-oriented communication      |              | .83      | 中立 — Neutral<br>同意 — Agree<br>非常同意 — Strongly agree           |                                              |

**Note.** s-EMBU = short-form Egnä Minnen av Barndoms Uppfostran; PERS-C = Chinese version of the Parent Emotion Regulation Scale. Chinese was the language of administration; English translations of response options are provided for readability. Item counts refer to the scored items used for the modeled dimensions. Internal consistency coefficients were calculated using the available questionnaire responses in the current sample.

## S2. Representative administered items

Table S2 provides one representative administered item for each modeled dimension.

**Table S2. Representative Chinese items with English translations.**

| Instrument                         | Dimension                           | Representative administered item<br>(Chinese and English translation)                                                                                                        |
|------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| s-EMBU                             | Emotional warmth                    | 我会夸奖我的孩子。<br>I praise my child.                                                                                                                                              |
| s-EMBU                             | Overprotection                      | 我会干涉孩子所做的一切。<br>I interfere with everything my child does.                                                                                                                   |
| s-EMBU                             | Rejection                           | 即使是小错，我也会严惩孩子。<br>I punish my child severely even for minor mistakes.                                                                                                        |
| PERS-C                             | Emotional attention                 | 我关注我孩子的情绪并试着去理解他们。<br>I pay attention to my child's emotions and try to understand them.                                                                                     |
| PERS-C                             | Emotional avoidance                 | 如果有可能，我会消除孩子的所有负面情绪。<br>If possible, I would eliminate all of my child's negative emotions.                                                                                  |
| PERS-C                             | Emotional acceptance                | 当我的孩子紧张时，我不是立即采取行动，而是给他时间让他自己冷静下来并解决问题。<br>When my child is distressed, I do not act immediately; instead, I give them time to calm down and solve the problem on their own. |
| PERS-C                             | Emotional control                   | 当我的孩子紧张时，我能够保持冷静并帮助他/她面对这种情况。<br>When my child is distressed, I am able to remain calm and help them face the situation.                                                     |
| Parent-child communication quality | Relationship-oriented communication | 孩子与我的交流让彼此更加亲密。<br>Communication between my child and me makes us feel closer.                                                                                               |
| Parent-child communication quality | Problem-oriented communication      | 孩子与我的讨论有助于解决问题。<br>Discussions between my child and me help solve problems.                                                                                                  |

**Note.** s-EMBU = short-form Egna Minnen av Barndoms Uppfostran; PERS-C = Chinese version of the Parent Emotion Regulation Scale. Chinese items are shown as administered; English translations were prepared for this supplement and were not administered to participants.

### S3. Prediction uncertainty and target correspondence

Tables S3A and S3B report paired comparisons between each final audio model and its corresponding non-informative baseline using the saved out-of-fold predictions, together with Pearson correlations between out-of-fold predictions and observed questionnaire scores. Target-specific sample sizes reflect the availability of valid questionnaire scores for each outcome.

**Table S3A. Out-of-fold MAE estimates and uncertainty.**

| Dimension and comparison                                        | n  | Audio MAE | Baseline MAE | MAE reduction (%) | 95% CI for MAE difference |
|-----------------------------------------------------------------|----|-----------|--------------|-------------------|---------------------------|
| Relationship-oriented communication<br>SVR-RBF vs mean baseline | 62 | 0.348     | 0.368        | +5.43             | [-0.0201, 0.0584]         |
| Problem-oriented communication<br>SVR-linear vs median baseline | 62 | 0.222     | 0.227        | +2.20             | [-0.0188, 0.0288]         |
| Emotional warmth<br>Random forest vs mean baseline              | 65 | 0.368     | 0.372        | +1.08             | [-0.0383, 0.0453]         |
| Overprotection<br>XGBoost vs median baseline                    | 65 | 0.446     | 0.433        | -3.00             | [-0.0699, 0.0442]         |
| Rejection<br>SVR-linear vs median baseline                      | 65 | 0.436     | 0.473        | +7.82             | [-0.0211, 0.0916]         |
| Emotional attention<br>SVR-linear vs mean baseline              | 62 | 0.379     | 0.420        | +9.76             | [-0.0107, 0.0919]         |
| Emotional avoidance<br>KNN vs median baseline                   | 62 | 0.715     | 0.724        | +1.24             | [-0.0602, 0.0759]         |
| Emotional acceptance<br>HistGB vs median baseline               | 62 | 0.493     | 0.492        | -0.20             | [-0.0017, 0.0001]         |
| Emotional control<br>Ridge vs mean baseline                     | 62 | 0.559     | 0.660        | +15.30            | [-0.0128, 0.2127]         |

*Note.* *n* denotes the number of families with available data for each dimension (overall sample: *N* = 66). Positive MAE differences favor the audio model; 95% CIs were estimated using 50,000 paired family-level bootstrap resamples.

**Table S3B. Pearson correlations between out-of-fold predictions and observed scores.**

| Dimension                           | n  | Pearson's <i>r</i> | 95% family-level bootstrap CI |
|-------------------------------------|----|--------------------|-------------------------------|
| Relationship-oriented communication | 62 | 0.226              | [0.0005, 0.4342]              |
| Problem-oriented communication      | 62 | 0.213              | [0.0145, 0.3978]              |
| Emotional warmth                    | 65 | 0.205              | [-0.0239, 0.4277]             |
| Overprotection                      | 65 | 0.055              | [-0.1789, 0.3088]             |
| Rejection                           | 65 | 0.339              | [0.1353, 0.5419]              |
| Emotional attention                 | 62 | 0.229              | [-0.0137, 0.4404]             |
| Emotional avoidance                 | 62 | 0.105              | [-0.1833, 0.3824]             |
| Emotional acceptance                | 62 | -0.227             | [-0.4789, 0.0522]             |
| Emotional control                   | 62 | 0.474              | [0.2614, 0.6520]              |

*Note.* Pearson's *r* was calculated from out-of-fold predictions; 95% CIs were estimated using 50,000 family-level bootstrap resamples.

#### **S4. Copyright and reuse note**

The questionnaire instruments and their Chinese adaptations originate from the cited publications and may be subject to copyright or other reuse restrictions. This supplement therefore reports measure structure and a limited number of representative items for scholarly transparency, rather than redistributing complete instruments. Researchers seeking to reuse complete questionnaires should consult the original sources and obtain any permissions required by the relevant rights holders.

## References

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